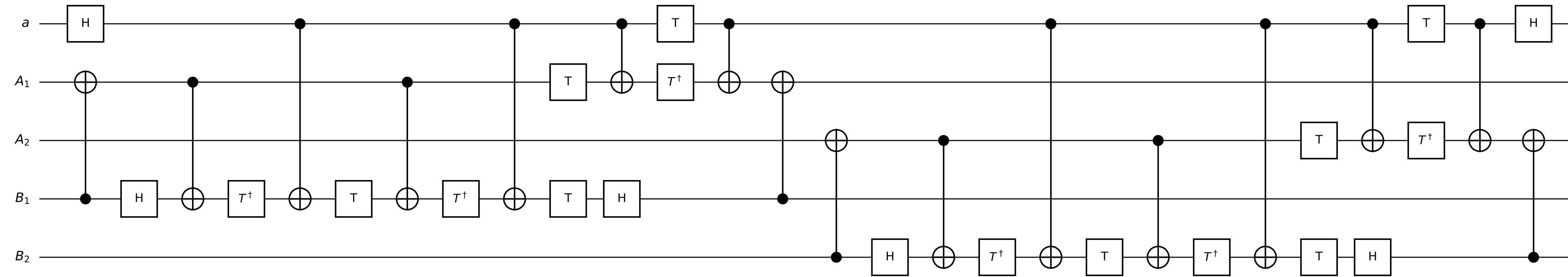


Step 3: textbook two-Controlled-SWAP decomposition (16 CNOTs)



A two-qubit replacement depolarizing channel D_q is applied after every CNOT.